

2. [Maximum mark: 31]

This question asks you to find the probability of graphs of randomly generated quadratic functions having a specified number of x -intercepts.

In parts (a) – (f), consider quadratic functions, $f(x) = ax^2 + bx + c$, whose coefficients, a , b and c , are randomly generated in turn by rolling an unbiased six-sided die three times and reading off the value shown on the uppermost face of the die.

For example, rolling a 2, 3 and 5 in turn generates the quadratic function $f(x) = 2x^2 + 3x + 5$.

- (a) Explain why there are 216 possible quadratic functions that can be generated using this method. [1]

- (b) The set of coefficients, $a = 1$, $b = 4$ and $c = 4$, is randomly generated to form the quadratic function $f(x) = x^2 + 4x + 4$.

Verify that this graph of f has only one x -intercept. [2]

- (c) By considering the discriminant, or otherwise, show that the probability of the graph of such a randomly generated quadratic function having only one x -intercept is $\frac{5}{216}$. [6]

Now consider randomly generated quadratic functions whose corresponding graphs have two **distinct** x -intercepts.

- (d) By considering the discriminant, determine the set of possible values of ac . [3]

- (e) (i) For the case where $ac = 1$, show that there are four quadratic functions whose corresponding graphs have two distinct x -intercepts. [1]

- (ii) For the case where $ac = 2$, show that there are eight quadratic functions whose corresponding graphs have two distinct x -intercepts. [2]

Let p be the probability of the graph of such a randomly generated quadratic function having two distinct x -intercepts.

- (f) Using the approach started in part (e), or otherwise, find the value of p . [6]

(This question continues on the following page)

(Question 2 continued)

In parts (g) and (h), consider a randomly generated quadratic function, $f(x) = x^2 + 2Zx + 1$, where the continuous random variable $Z \sim N(0, 1)$.

- (g) Find the probability that the graph of f has two x -intercepts. [3]

The continuous random variables, X_1 and X_2 , represent the x -intercepts of the graph of f where $X_1 = -Z - \sqrt{Z^2 - 1}$ and $X_2 = -Z + \sqrt{Z^2 - 1}$.

- (h) Given that the graph of f has two x -intercepts, X_1 and X_2 , find the probability that both X_1 and X_2 are greater than 0.5. [7]
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